

Trainings ▾

Preparatory Steps

WARNING

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

WARNING

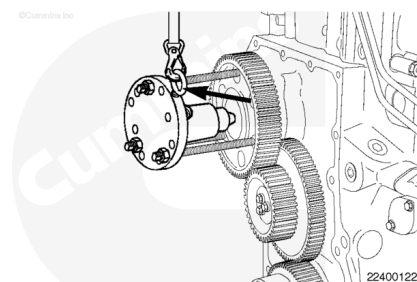
Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Remove rocker lever covers from cylinder 2 and 5. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/89/89-003-011.html>)
- Push the barring device shaft in and turn **counterclockwise** until the "I-6TC" mark on the damper is aligned with the pointer attached to the front of the block.
- For cylinders 2 and 5, identify the injector push rod which is higher in relation to the top of the rocker lever housing. If the injector push rod is higher on cylinder number 2, cylinder number 1 is on its compression stroke. If **not**, rotate the crankshaft one full revolution to place number 1 cylinder on its compression stroke.
- Drain the oil from the lubricating oil pan. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/89/89-007-037.html>)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>)
- Remove the cylinder block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>)
- Remove the flywheel housing and all related components. Refer to Procedure 016-006 in Section 16. (</qs3/pubsys2/xml/en/procedures/89/89-016-006.html>)

Remove

CAUTION

Do not change crankshaft position while the camshaft is removed.



Make sure the gear teeth with the “B” marks on the cam gear and the smaller idler gear are in mesh and the keyway in the crankshaft is at 6 o'clock.

Use the Camshaft Gear Puller Kits, Part Numbers 3164375, and 3164925, to remove camshaft gear.

Note : Both kit part numbers are necessary when removing the camshaft gear.

Attach the puller to the camshaft gear.

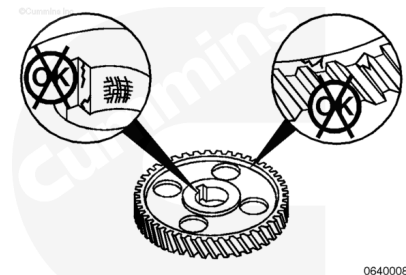
Remove the camshaft gear.

Remove the puller.

Inspect for Reuse

Inspect the camshaft gear for cracks or chipped or broken teeth.

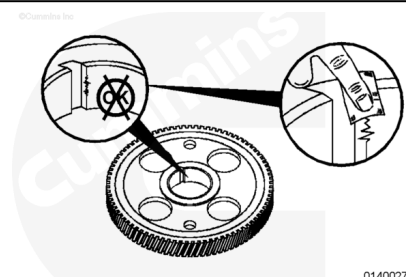
Inspect the bore of the gear for fretting or burrs.



Inspect the gear keyway for burrs.

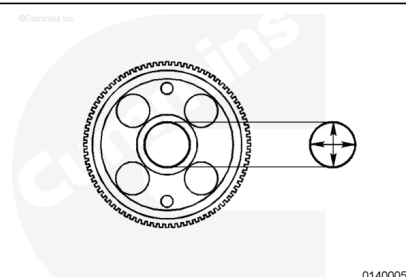
Remove any burrs with fine crocus cloth.

If the keyway is damaged or it is **not** possible to remove the burrs with fine crocus cloth, the gear **must** be replaced.



Measure the inside diameter of the camshaft gear.

Camshaft Gear Inside Diameter		
mm		in
80.00	MIN	3.150
80.02	MAX	3.150



The gear **must** be replaced if the inside diameter is **not** within specifications.

Install

⚠ WARNING ⚠

To reduce the possibility of severe burns, wear protective gloves when installing the heated ring gear.

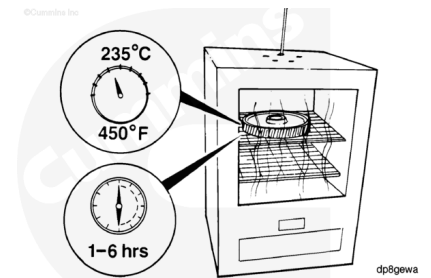
⚠ CAUTION ⚠

Do not exceed the specified time or the temperature. Damage to the gear and the gear teeth will result.

⚠ CAUTION ⚠

Do not attempt to install the gear without using heat. Damage to the camshaft and camshaft gear can result.

Use an oven and adjust the heat to 235°C [450°F]. Heat the gear for a minimum of 1 hour, but no more than a maximum of 6 hours. The inside diameter of the gear will become larger and will simplify installation.



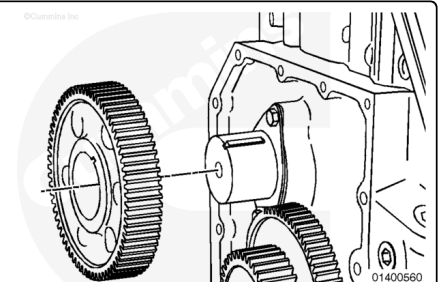
⚠ CAUTION ⚠

Allow the gear to cool slowly. Do not use water or oil to reduce the cooling time. This will cause the gear to crack.

If removed, install the camshaft key into the slot on the camshaft.

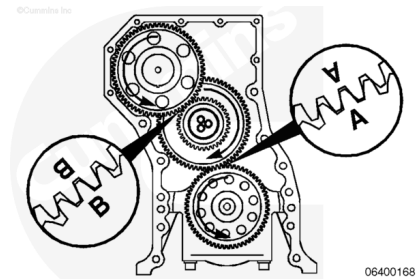
Note : There is **only** a straight key option. It is **not** necessary to check/adjust static injection timing on the QSK23 engine.

The timing mark on the camshaft gear **must** be visible from the front of the gear after it is installed on the camshaft.



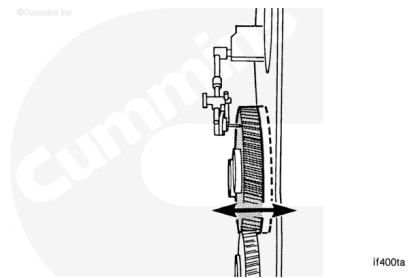
Remove the gear from the oven and install it on the camshaft.
The keyway in the gear **must** be aligned with the key in the camshaft and the “B” marks on the cam gear and the smaller idler gear **must** also be aligned.

After installing the gear and camshaft, double check that the "A" and "B" marks are lined up on the camshaft compound idler gear.



Use a dial indicator and measure the camshaft end clearance.

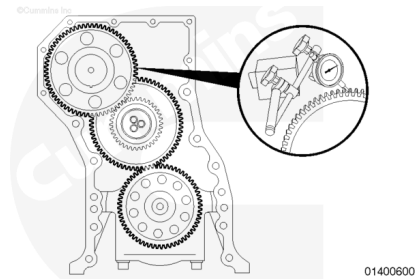
Camshaft End Clearance		
mm		in
0.05	MIN	0.002
0.20	MAX	0.008



If the clearance is too small, check for foreign material or a piece of gasket between the thrust plate and the block.

Use a dial indicator to measure the camshaft gear backlash.

Camshaft Gear Backlash		
mm		in
0.145	MIN	0.006
0.380	MAX	0.015



Finishing Steps

- Install the flywheel housing and all related components. Refer to Procedure 016-006 in Section 16. (</qs3/pubsys2/xml/en/procedures/89/89-016-006.html>)
- Install the cylinder block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>)

- Install the rocker lever covers for cylinders 2 and 5. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/89/89-003-011.html)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/89/89-007-037.html)

Last Modified: 27-Jun-2019
